



Fig. 2: 3D model slicing illustration (Source: http://en.wikipedia.org/wiki/3D_printing)

The 3D printer then processes these instructions and creates each layer as per specifications. As the layers are made, they are combined together in a seamless manner and, finally, a 3D object is created. Autodesk is one of the leading players involved in creating software for 3D printing.

While conventional manufacturing involves subtractive processes, that is, removal of material by chiselling, cutting, etc, 3D printing is based on an additive process, where successive layers of material build different shapes. Objects are created by means of a sequential layering process. 3D printing technology is being used to make jewellery and footwear, in industrial design, architecture, engineering and construction, as well as in the automotive, aerospace, dental/medical, education, geographic information systems and several other industries.

Several companies are using 3D printing. For example, the global sports shoes manufacturer, Nike, uses 3D printing to develop multi-coloured prototypes of shoes. This has brought down both the cost and time spent in prototype development for Nike, considerably.

Let us understand the modelling part first. Additive manufacturing works on virtual blueprints from CAD or animation modelling software, and 'slices' them into digital cross-sections so that they serve as a guideline for the printing to be done. Depending on the

printer used, a binding material is deposited on the build bed or platform until the material/binder layering is complete and the final 3D model has been 'printed.' It is a WYSIWYG process where the virtual and the physical models are mostly similar. The STL file format serves as a standard data interface between CAD software and the machines. It estimates the shape of a part or assembly using triangular

facets. Smaller facets produce a higher quality surface. PLY is a scanner generated input file format, and VRML (or WRL) files are frequently used as input for 3D printing technologies that are able to print in full colour.

The printing machine understands the design via a .stl file and lays down consecutive layers of powder, liquid or sheet material, i.e., whatever is needed to make the model from a number of cross-sections. Thereafter, these layers, which are in line with the virtual cross-sections from the CAD model, are either connected together or automatically fused to make the desired shape. The key advantage of this technique is its ability to make just about any contour or form.

The printer resolution defines the layer thickness and X-Y resolution in dpi (dots per inch), or micrometres.



Fig. 3: A 3D sandal printed out of nylon (Source: Chemistry & Industry April 25, 2011)

On an average, the thickness is about 0.1 mm. However, some printers such as the Objet Connex series and 3D Systems' ProJet series can print layers as thin as .016 micrometres.

The time taken for model construction using contemporary methods ranges from a few hours to a few days. The actual time taken would depend on the method used, and the size and complexity of the model. Additive systems can bring down this time to a few hours depending on the nature of machine used, and the dimensions and quantity of the models being created simultaneously.

While conventional techniques, such as injection moulding, cost less for manufacturing polymer products in large numbers, 3D printing is speedier, more flexible and also cheaper when the number of units is low. The greatest advantage is that design and concept development teams have the freedom and power to produce parts and concept models using their own desktop-size printer.

Choosing a 3D printer

To start with, a firm needs to determine whether a 3D printer is really required or not, i.e., whether using a service provider would be better or would owning a 3D printer lead to higher productivity, lower costs and quicker delivery. 3D printing is a solution that takes care of all the modelling and prototyping needs of an organisation. Another purpose that 3D printing serves is that of confidentiality – especially with a new design, since blueprints or CAD designs do not go out of the company at all. How you intend using the 3D printer will determine what type you buy. For instance, if you plan to use the 3D printer in the concept development stage for product design and for design validation, you should go in for a high-speed full-colour printer with labelling options. Test-driving the printer's control software will tell you how easy or difficult it is to send .stl files to the 3D printer.

Apart from commercial printers,